Reflection and Traceability Report on Software Engineering

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[Reflection is an important component of getting the full benefits from a learning experience. Besides the intrinsic benefits of reflection, this document will be used to help the TAs grade how well your team responded to feedback. Therefore, traceability between Revision 0 and Revision 1 is and important part of the reflection exercise. In addition, several CEAB (Canadian Engineering Accreditation Board) Learning Outcomes (LOs) will be assessed based on your reflections. —TPLT]

1 Changes in Response to Feedback

[Summarize the changes made over the course of the project in response to feedback from TAs, the instructor, teammates, other teams, the project supervisor (if present), and from user testers. —TPLT]

[For those teams with an external supervisor, please highlight how the feedback from the supervisor shaped your project. In particular, you should highlight the supervisor's response to your Rev 0 demonstration to them. —TPLT]

[Version control can make the summary relatively easy, if you used issues and meaningful commits. If your feedback is in an issue, and you responded in the issue tracker, you can point to the issue as part of explaining your changes. If addressing the issue required changes to code or documentation, you can point to the specific commit that made the changes. Although the links are helpful for the details, you should include a label for each item of feedback so that the reader has an idea of what each item is about without the need to click on everything to find out. —TPLT]

[If you were not organized with your commits, traceability between feedback and commits will not be feasible to capture after the fact. You will instead need to spend time writing down a summary of the changes made in response to each item of feedback. —TPLT]

[You should address EVERY item of feedback. A table or itemized list is recommended. You should record every item of feedback, along with the source of that feedback and the change you made in response to that feedback. The response can be a change to your documentation, code, or development process. The response can also be the reason why no changes were made in response to the feedback. To make this information manageable, you will record the feedback and response separately for each deliverable in the sections that follow. —TPLT]

[If the feedback is general or incomplete, the TA (or instructor) will not be able to grade your response to feedback. In that case your grade on this document, and likely the Revision 1 versions of the other documents will be low. —TPLT]

1.1 Problem Statement, Goals, and Development Plan

Table 1 summarizes the feedback received from our peers, the TA, the instructor and other sources. The table includes links to the related issues and pull requests, along with a summary of our response/changes.

Table 1: Feedback Responses and Traceability for Problem Statement, Goals, Development Plan

Feedback Source	**Summary**	**Changes Made**	**Related Issues**
Peer	Learning outcome extraction lacks a measurable success criteria	Added measurable success criteria to the learning outcome extraction	IS#32, Commit#2a966de
Peer	Word document support is not explicitly addressed in the schedule	Clarified that initially only PDF support will be implemented and Word document support will be documented as future work.	IS#33, Commit#f07d482

Table 1: Feedback Responses and Traceability for Problem Statement, Goals, Development Plan

Feedback Source	Summary	Changes Made	Related Issues
TA	Need more details in the team charter: How is "repeatedly misses a commitment" defined? What's the exact amount? What would work balancing look like?	Updated team charter to include examples and more details.	IS#94, PR#117
Peer	The PoC fallback could leave the CP-SAT constraint risk unresolved	Prioritized a reduced CP-SAT model and clarified that any scoring fallback must document the unresolved constraints and evaluate them before full implementation.	IS#35
Peer	CEAB validation is not clearly demonstrated in the PoC	Added one representative CEAB graduate-attribute check to the PoC and clarified that broader CEAB coverage will be evaluated afterward.	IS#34
TA	The PoC plan is too ambitious and does not clearly prioritize its main risks	Made reliable data extraction and alternative scheduling dates the primary PoC focus, while limiting policy, CEAB, explanation, and other features to small demonstrations or follow-up work.	IS#93
Other	summary of feedback	summary of changes made	IS#, PR#, ...

1.2 SRS and Hazard Analysis

The SRS and Hazard Analysis deliverables are closely related, so we will summarize the feedback received for these two deliverables together in the sections that follow.

1.2.1 SRS

Table 2 summarizes the feedback received from our peers, the TA, the instructor and other sources. The table includes links to the related issues and pull requests, along with a summary of our response/changes.

Table 2: Feedback Responses and Traceability for SRS

Feedback Source	Summary	Changes Made	Related Issues
Peer	Session timeout protection is not formally specified in the SRS and Hazard Analysis	Added a requirement specifying that authenticated sessions should expire after a defined period of inactivity in the SRS and updated the Hazard Analysis accordingly	IS#86, PR#118

Table 2: Feedback Responses and Traceability for SRS

Feedback Source	Summary	Changes Made	Related Issues
Peer	Model changes could reduce extraction accuracy without a formal regression requirement	Added a requirement to rerun the representative accuracy dataset and confirm the thresholds before deploying a new or updated model.	IS#84
Peer	CSV export did not specify protection against spreadsheet formula interpretation	Updated the CSV export requirement to escape values beginning with formula characters.	IS#85
TA	summary of feedback	summary of changes made	IS#, PR#, ...
Instructor	summary of feedback	summary of changes made	IS#, PR#, ...
Other	summary of feedback	summary of changes made	IS#, PR#, ...

1.2.2 Hazard Analysis

Table 3 summarizes the feedback received from our peers, the TA, the instructor and other sources. The table includes links to the related issues and pull requests, along with a summary of our response/changes.

Table 3: Feedback Responses and Traceability for Hazard Analysis

Feedback Source	Summary	Changes Made	Related Issues
Peer	Backup and recovery safeguards are not clearly defined	Clarified H2-1 by assigning backups and recovery to CAS IT, requiring the project team to document application data and verify representative restores, and adding a backup and restore requirement to the roadmap.	IS#83, PR#123, ...
Peer	CSV formula injection prevention was not included in the safety requirements	Added SSR_CSVExport to formalize escaping dangerous spreadsheet formula prefixes and testing the resulting export.	IS#85
Peer	Model changes lacked a defined regression testing requirement	Added SSR_ModelChangeTest to require accuracy testing and recording the model version before deployment.	IS#84
TA	summary of feedback	summary of changes made	IS#, PR#, ...
Instructor	summary of feedback	summary of changes made	IS#, PR#, ...
Other	summary of feedback	summary of changes made	IS#, PR#, ...

1.3 Design and Design Documentation

The design and design documentation deliverables include the software architecture and detailed design. We will summarize the feedback received for each of these deliverables separately in the sections that follow.

1.3.1 Software Architecture

Table 4 summarizes the feedback received from our peers, the TA, the instructor and other sources. The table includes links to the related issues and pull requests, along with a summary of our response/changes.

Table 4: Feedback Responses and Traceability for Software Architecture

Feedback Source	Summary	Changes Made	Related Issues
Peer	summary of feedback	summary of changes made	IS#, PR#, ...
TA	summary of feedback	summary of changes made	IS#, PR#, ...
Instructor	summary of feedback	summary of changes made	IS#, PR#, ...
Other	summary of feedback	summary of changes made	IS#, PR#, ...

1.3.2 Detailed Design

Table 5 summarizes the feedback received from our peers, the TA, the instructor and other sources. The table includes links to the related issues and pull requests, along with a summary of our response/changes.

Table 5: Feedback Responses and Traceability for Detailed Design

Feedback Source	Summary	Changes Made	Related Issues
Peer	summary of feedback	summary of changes made	IS#, PR#, ...
TA	summary of feedback	summary of changes made	IS#, PR#, ...
Instructor	summary of feedback	summary of changes made	IS#, PR#, ...
Other	summary of feedback	summary of changes made	IS#, PR#, ...

1.4 VnV Plan and Report

The VnV Plan and Report deliverables are closely related, so we will summarize the feedback received for these two deliverables together in the sections that follow.

1.4.1 VnV Plan

Table 6 summarizes the feedback received from our peers, the TA, the instructor and other sources. The table includes links to the related issues and pull requests, along with a summary of our response/changes.

Table 6: Feedback Responses and Traceability for VnV Plan

Feedback Source	Summary	Changes Made	Related Issues
Peer	summary of feedback	summary of changes made	IS#, PR#, ...
TA	summary of feedback	summary of changes made	IS#, PR#, ...
Instructor	summary of feedback	summary of changes made	IS#, PR#, ...
Other	summary of feedback	summary of changes made	IS#, PR#, ...

1.4.2 VnV Report

Table 7 summarizes the feedback received from our peers, the TA, the instructor and other sources. The table includes links to the related issues and pull requests, along with a summary of our response/changes.

Table 7: Feedback Responses and Traceability for VnV Report

Feedback Source	Summary	Changes Made	Related Issues
Peer	summary of feedback	summary of changes made	IS#, PR#, ...
TA	summary of feedback	summary of changes made	IS#, PR#, ...

Table 7: Feedback Responses and Traceability for VnV Report

Feedback Source	Summary	Changes Made	Related Issues
Instructor	summary of feedback	summary of changes made	IS#, PR#, ...
Other	summary of feedback	summary of changes made	IS#, PR#, ...

2 Custom Documents (CDs)

[Summarize the CDs that were tackled by this project. CDs can include usability testing, code walkthroughs, user documentation, formal proof, GenderMag personas, Design Thinking, etc. CDs should have already been approved by the course instructor as included in your problem statement. The CDs should match what is documented in TeamComposition.csv. —TPLT]

[Provide links to the documentation of the CDs —TPLT]

3 Design Iteration (LO10 (Prototypes))

[Explain how you arrived at your final design and implementation. How did the design evolve from the first version to the final version? —TPLT]

[Don't just say what you changed, say why you changed it. The needs of the client should be part of the explanation. For example, if you made changes in response to usability testing, explain what the testing found and what changes it led to. —TPLT]

4 Design Decisions (LO11) (Justifies)

[Reflect and justify your design decisions. How did limitations, assumptions, and constraints influence your decisions? Discuss each of these separately. —TPLT]

5 Economic Considerations (LO19) (Economic)

[Is there a market for your product? What would be involved in marketing your product? What is your estimate of the cost to produce a version that you could sell? What would you charge for your product? How many units would you have to sell to make money? If your product isn't something that would be sold, like an open source project, how would you go about attracting users? How many potential users currently exist? —TPLT]

6 Reflection on Project Management (LO20) (Manages)

[This question focuses on processes and tools used for project management. —TPLT]

6.1 How Does Your Project Management Compare to Your Development Plan

[Did you follow your Development plan, with respect to the team meeting plan, team communication plan, team member roles and workflow plan. Did you use the technology you planned on using? —TPLT]

6.2 What Went Well?

[What went well for your project management in terms of processes and technology? —TPLT]

6.3 What Went Wrong?

[What went wrong in terms of processes and technology? —TPLT]

6.4 What Would you Do Differently Next Time?

[What will you do differently for your next project? —TPLT]

7 Ethics and Equity (LO18) (Equity)

[Describe how your team applied the principle of equity and universal design to ensure equitable treatment of all stakeholders. —TPLT]

8 Reflection on Capstone

[This question focuses on what you learned during the course of the capstone project. —TPLT]

8.1 Which Courses Were Relevant

[Which of the courses you have taken were relevant for the capstone project? —TPLT]

8.2 Knowledge/Skills Outside of Courses

[What skills/knowledge did you need to acquire for your capstone project that was outside the courses you took? —TPLT]